

KAUNAS UNIVERSITY OF TECHNOLOGY INFORMATICS
FACULTY

INTRODUCTION TO ARTIFICIAL INTELLIGENCE
LAB3

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1. Initial Dataset Description

Dataset: heart.csv the dataset is used for building a binary classification model (same dataset as in lab 1).

Target attribute: target (indicates the presence or absence of heart disease).

Input attributes: 13 distinct features representing patient medical data and test results (e.g., age, cholesterol levels, resting blood pressure, max heart rate).

2. A List of Possible Dataset Rearrangements with Comments

Feature Scaling / Standardization (Implemented): Applying a scaler to transform numerical data so it has a mean of 0 and a standard deviation of 1.

This is crucial because features with larger numerical ranges can disproportionately dominate the network's weight updates. Scaling ensures all features contribute equally and helps the gradient descent converge faster.

Categorical Encoding: Converting categorical variables into binary dummy variables (One-Hot Encoding).

Prevents the neural network from incorrectly assuming that categorical numbers have a mathematical order or hierarchy.

3. Architecture of Baseline ANN

The initial model was built to establish a baseline prediction performance.

Input Layer: 13 neurons (matching the 13 input features).

Hidden Layers: 1 hidden layer containing 8 neurons.

Output Layer: 1 neuron (for binary classification).

Parameters:

Activation function: logistic (Sigmoid)

Initial learning rate: 0.001

Maximum iterations: 1000

4. Results of 10-Fold Cross Validation Experiments (Baseline)

The model was evaluated using 10-fold cross-validation to ensure reliable, averaged metrics.

Accuracy at each fold:

Averaged Performance:

```
--- BASELINE MODEL ---  
Accuracy at each fold: [82.52 81.55 89.32 86.41 86.41 87.25 90.2 88.24 86.27 78.43]  
Averaged Performance: 85.66%
```

5. Description of Measures Taken to Increase ANN Performance

To achieve the goal of increasing performance by at least 5%, the following four measures were simultaneously applied using a data pipeline:

1. **Rearranged Dataset:** Applied StandardScaler() within the pipeline to normalize the dataset dynamically during each fold of the cross-validation, preventing data leakage.
2. **Changed ANN Structure:** Deepened the architecture from a single hidden layer to two hidden layers. The new structure uses 16 neurons in the first hidden layer and 8 neurons in the second.
3. **Changed Activation Function:** Switched from logistic to relu (Rectified Linear Unit) to mitigate potential vanishing gradients and allow the network to learn more complex, non-linear relationships.
4. **Changed Learning Rate:** Increased the initial learning rate from 0.001 to 0.01 to accelerate the weight optimization process.

Improved Model 10-Fold Cross Validation Results:

Accuracy at each fold:

Averaged Performance:

Total Improvement:

```
--- IMPROVED MODEL ---  
Accuracy at each fold: [ 97.09 99.03 100. 100. 100. 99.02 100. 99.02 100. 100. ]  
Averaged Performance: 99.42%  
  
Total Improvement: 13.75%
```

6. Conclusions

The initial neural network provided a foundational baseline, but struggled to capture the full complexity of the heart.csv dataset due to unscaled inputs and a shallow architecture. By systematically applying data standardization, expanding the network depth (to 16 and 8 neurons), adopting the ReLU activation function, and increasing the learning rate, the model's predictive power was significantly enhanced. The 10-fold cross-validation confirmed that these combined measures successfully generalized across the dataset, yielding a performance improvement that met the laboratory requirements.